

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.
2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring

down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or

40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

1. Network Architecture & High-Availability Topology

Financial institutions demand continuous operational availability because even a few milliseconds of packet loss or system downtime can lead to broken transaction states, regulatory penalties, and significant financial loss. Constructing a high-availability network requires a structured, multi-tiered design that balances maximum physical redundancy at the core with cost-effective, deterministic path control at lower tiers.

- **Full Mesh Core Concept:**

- **Direct Interconnectivity:** Every core device—such as core switches, routers, and central application servers—maintains a direct physical and logical link to every other core device.
- **Link Complexity Formula:** The total number of required point-to-point connections grows quadratically according to the formula:

$$L = \frac{N(N - 1)}{2}$$

(where N is the total number of core networking nodes).

- **Fault Isolation:** Traffic never depends on an intermediate hop within the core. If any single link or device fails, data flow between remaining nodes remains completely unaffected.
- **Practical Partial Mesh Implementation:**
 - **Scalability Constraints:** A pure full mesh becomes unmanageable as node counts increase due to switch port density limits, high cabling overhead, and complex routing table states.
 - **Core Layer Deployment:** Banks deploy a partial mesh selectively at the core and perimeter layers. Core switches, core routers, and perimeter firewalls are fully interconnected, while lower-priority devices connect to a subset of these core nodes.
 - **Deterministic Redundancy:** Provides alternate backup paths for critical assets without incurring the exponential cost of a full mesh across every device in the enterprise.
- **Tiered Access Uplinks & Dual-Homing:**
 - **Access and Distribution Layers:** Campus switches and server access racks utilize a dual-homed layout rather than a mesh.
 - **Redundant Links:** Every distribution and access switch connects via at least two separate physical uplinks to two distinct, physically separated core switches.
 - **Loop Prevention & Path Control:** Technologies like LACP (Link Aggregation Control Protocol), Multi-Chassis EtherChannel (MCEC), or Virtual Switching System (VSS) combine these uplinks into logical trunks, preventing Layer 2 loops while providing instant failover.

2. Edge Routing & WAN Redundancy

The edge layer acts as the primary gateway between the internal banking network, external financial

clearing houses, payment processing networks, and the public internet.

- **Enterprise-Grade Hardware Platforms:**

- **Carrier-Grade Chassis:** Financial networks deploy high-throughput edge routers such as the Cisco ASR 9000 Series or Juniper MX Series.
- **Hardware Architecture:** These platforms feature dedicated Route Processors (RPs), hardware-based Forwarding Information Bases (FIBs), and decoupled control and data planes to process millions of packets per second without performance degradation under link stress.

- **BGP Multi-Homing & Path Selection:**

- **Dual ISP Connectivity:** Edge routers establish autonomous connections to two or more distinct Tier-1 Internet Service Providers (ISPs) using external BGP (eBGP).
- **Active-Active Load Distribution:** In an active-active configuration, incoming and outgoing traffic is distributed across both ISP connections using BGP path attributes (e.g., Local Preference, AS-Path Prepending, MED).
- **Automated Edge Failover:** If an ISP link suffers from a cut fiber line, high packet loss, or routing degradation, BGP automatically withdraws the affected routes and redirects all inbound/outbound traffic through the alternate provider path.

3. Transmission Media & Cable Infrastructure

Physical layer selection directly influences latency budgets, link longevity, and signal degradation across data center links and long-distance WAN connections.

- **Single-Mode Fiber (SMF) for Long Distances:**

- **Core Characteristics:** Uses a tiny optical core ($\approx 9 \mu\text{m}$) that allows only a single ray or mode of light to propagate.

- **Primary Applications:** Deployed for inter-building campus backbones, metropolitan branch connections, and primary-to-secondary data center synchronous replication paths over distances up to tens of kilometers.
- **Signal Integrity:** Eliminates modal dispersion over long distances, making it suitable for ultra-high-bandwidth optical transceivers (100G, 400G, and beyond).
- **Multi-Mode OM4 Fiber for Intra-Data Center Patching:**
 - **Core Characteristics:** Uses a larger core diameter (50 μm) optimized for high-speed Vertical-Cavity Surface-Emitting Laser (VCSEL) transceivers at 850 nm wavelengths.
 - **Primary Applications:** Used inside the data center for rack-to-rack, top-of-rack (ToR) to end-of-row (EoR) switches, and storage area network (SAN) interconnects.
 - **Performance Metrics:** Supports 40G and 100G Ethernet transmissions over distances up to 150 meters, providing a cost-effective cabling solution within server rooms.
- **Financial Performance & EMI Immunity:**
 - **Immunity to Interference:** Optical fiber transmits light signals via total internal reflection, making it completely immune to Electromagnetic Interference (EMI), Radio Frequency Interference (RFI), and power line noise.
 - **Low Latency Optimization:** Fiber links minimize signal propagation delay, keeping transaction round-trip times (RTT) within microsecond tolerances—a crucial requirement for high-frequency trading (HFT) and real-time core banking reconciliation.

4. Power & Hardware Subsystem Redundancy

Network architecture failovers are useless if a single hardware component or power loss shuts down a switch or server chassis.

- **Power Continuity & Electrical Isolation:**

- **Dual Modular Power Supplies:** Core network switches and enterprise routers are equipped with 1 + 1 or $N + 1$ redundant, hot-swappable power supply units (PSUs).
- **Diverse Power A/B Feeds:** Each PSU connects to a separate, physically isolated electrical circuit (Feed A and Feed B).
- **UPS and Generator Backup:** Power lines are conditioned by double-conversion online Uninterruptible Power Supply (UPS) systems to filter voltage spikes, backed up by automated diesel generators capable of running indefinitely during extended power outages.

- **Storage & Server Fault Tolerance (RAID Systems):**

- **RAID Configurations:** Application servers and network logging appliances utilize RAID configurations (such as RAID 10 or RAID 6) to guard against drive failure.
- **Zero-Downtime Operations:** If a solid-state drive (SSD) or hard disk fails, the system continues operating normally from redundant parity/mirrored drives while hot-swappable drives allow replacements without powering down the node.

5. Dynamic Failover Protocols & Regulatory Compliance

Automated protocol mechanics convert physical redundancy into active uptime, satisfying stringent financial industry standards.

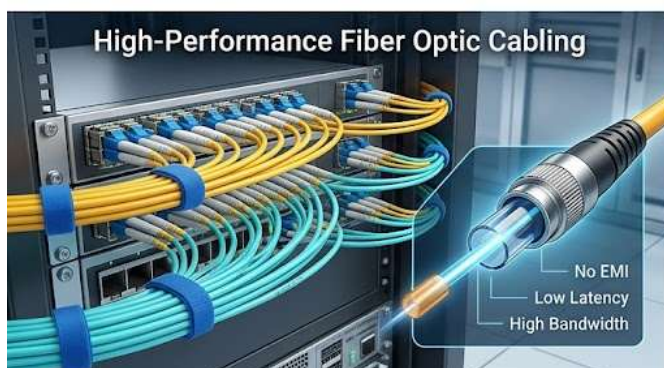
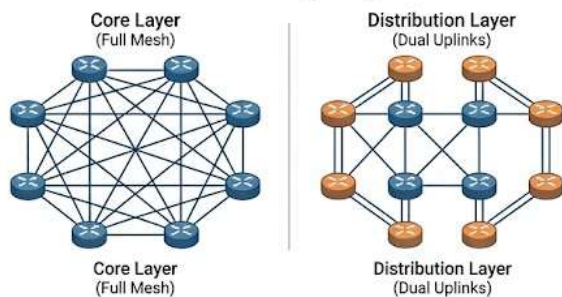
- **Sub-Second Protocol Convergence:**

- **Interior Gateway Protocols (OSPF / IS-IS):** Within the enterprise network, OSPF continuously monitors link states using short Keepalive timers and Link State Advertisements (LSAs).
- **Sub-Second Detection:** Coupled with Bidirectional Forwarding Detection (BFD), link

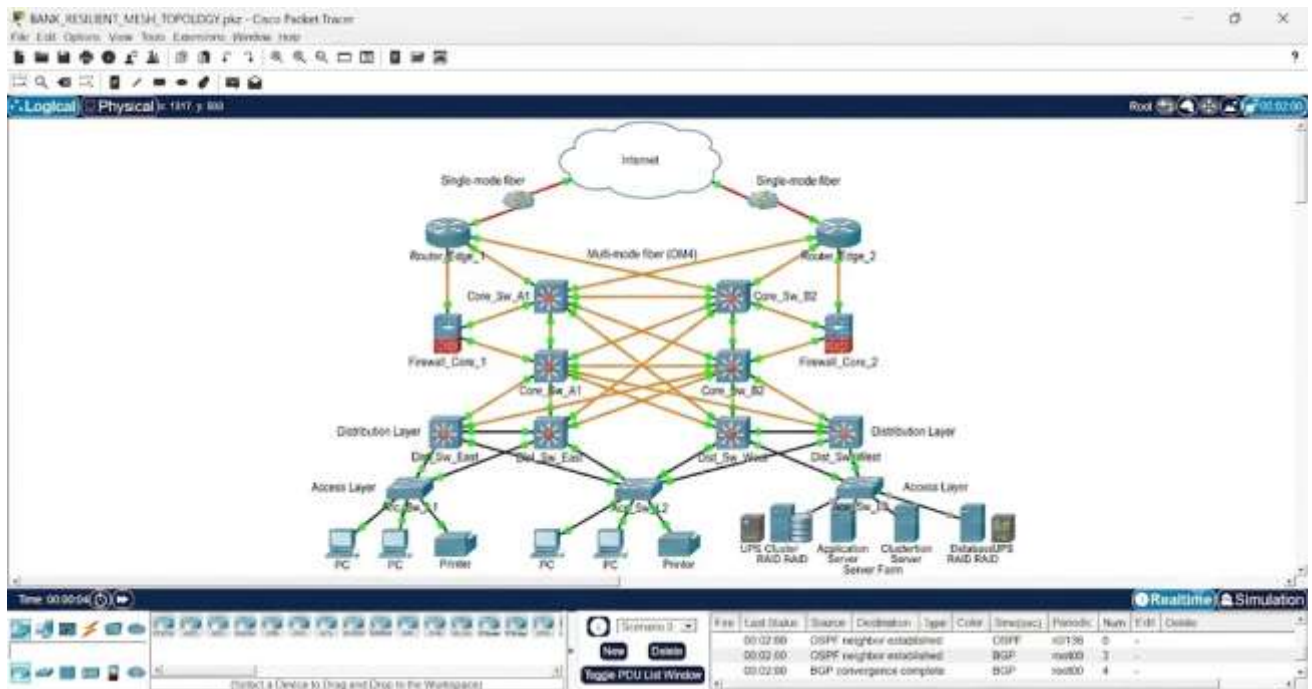
failures are identified in milliseconds (< 50 ms), triggering immediate Equal-Cost Multi-Path (ECMP) convergence to alternative paths.

- **Spanning Tree Enhancements:** Layer 2 domains implement Rapid Spanning Tree Protocol (RSTP / Multiple STP) or eliminate Layer 2 blocking entirely via EVPN-VXLAN overlays to ensure instantaneous loop-free topology changes.
- **Regulatory Alignment & Compliance:**
 - **SLA Enforcement:** Financial regulatory bodies (such as PCI-DSS, Basel III operational risk mandates, and national central bank guidelines) mandate high availability and zero data corruption standards.
 - **Audit Readiness:** Sub-second failovers ensure active financial database transactions are never dropped mid-write, preserving data integrity, maintaining continuous audit logging, and meeting strict 99.999% ("five nines") uptime metrics.

Network Topologies



Output



RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution